

Partial Derivatives and Gradients

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Derivatives of one variable ([Note 600](#)).
- **Leads to:** Jacobian and matrix gradients ([Note 602](#)); Lagrange multipliers, KKT and duality ([Note 620](#)).

1. Overview

Key point: For a function of many inputs, we take one derivative per input with the others held fixed; collected in a row, these partial derivatives form the gradient, an arrow that points straight uphill.

This Note follows Chapter 5 (Section 5.2) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

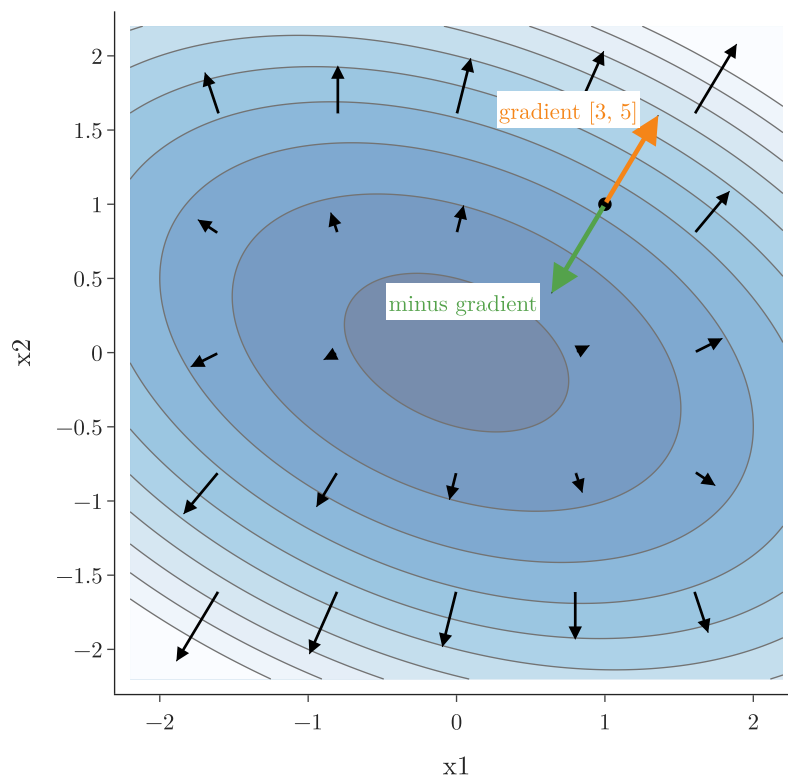


Figure 1: Contour map of $f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$ with gradient arrows: each crosses the contour lines at right angles and points uphill

A loss depends on many parameters at once. Figure 1 shows such a function of two inputs as

a contour map: each line joins points of equal height, and the lowest point is in the middle. At every point, the gradient is an arrow pointing in the steepest uphill direction. Its opposite, the green arrow, is the step gradient descent takes.

Earlier Notes already used these ideas:

- the partial derivative as a slope in one variable, from the [linear regression maths Note](#);
- the gradient as the vector of partial derivatives of the loss, from the [gradient descent Note](#).

This Note adds the definition as a limit, the geometry of the gradient, the rules for vectors and the chain rule with several variables. It builds on the [one-variable derivative of the derivatives of one variable Note](#).

2. Functions of several variables

Key point: $f : \mathbb{R}^n \rightarrow \mathbb{R}$ takes a vector of n numbers in and gives one number out, just as a loss takes all the parameters and gives one error.

A function of several variables takes a vector $\mathbf{x} = [x_1, \dots, x_n]^\top$ and returns one number:

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{x} \mapsto f(\mathbf{x})$$

Two examples we use throughout:

- the squared length of a vector, $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{x} = x_1^2 + x_2^2$, the dot product of $\mathbf{x}$ with itself (see the [dot product and cosine similarity Note](#));
- our running example, a tilted bowl like a loss surface:

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + 2x_2^2$$

At $(1, 1)$ its value is $1 + 1 + 2 = 4$.

3. Partial derivatives

Key point: A partial derivative changes one input by a tiny step, keeps all the others fixed, and measures the slope; it is an ordinary one-variable derivative.

3.1 The definition

Key point: Nudge only x_i by h , take the difference quotient, and let $h \rightarrow 0$.

The [linear regression maths Note](#) defined the **partial derivative** as the slope of a function of several variables in one variable, holding the others fixed. Written with the limit of the [derivatives of one variable Note](#):

1. **In words:** move only x_i by a step h , divide the change in f by h , and let h shrink to 0.

2. **Formula:**

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(\mathbf{x})}{h}$$

The curly ∂ (read “partial”) replaces d to say that the other variables are held fixed.

3. **Example:** for our bowl at $(1, 1)$ with $h = 0.1$,

$$\frac{f(1.1, 1) - f(1, 1)}{0.1} = \frac{1.21 + 1.1 + 2 - 4}{0.1} = 3.1$$

Smaller h brings this towards 3.

3.2 Computing partial derivatives

Key point: Treat every other variable as a constant and use the one-variable rules.

Because the other variables are frozen, they behave like numbers. All the rules of the [derivatives of one variable](#) [Note](#) apply unchanged.

1. **In words:** to differentiate with respect to x_1 , read x_2 as a fixed number, and the other way round.
2. **Formula:** for $f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$,

$$\frac{\partial f}{\partial x_1} = 2x_1 + x_2, \quad \frac{\partial f}{\partial x_2} = x_1 + 4x_2$$

In $\partial f / \partial x_1$, the term x_1x_2 is “ x_1 times a constant”, so its derivative is that constant x_2 ; the term $2x_2^2$ is a constant, so its derivative is 0.

3. **Example:** at $(1, 1)$: $\partial f / \partial x_1 = 2 + 1 = 3$ and $\partial f / \partial x_2 = 1 + 4 = 5$.

The chain rule works the same way inside a partial derivative. For $g(x, y) = (x + y^2)^2$ at $(1, 2)$, where the inside is $x + y^2 = 5$:

$$\frac{\partial g}{\partial x} = 2(x + y^2) \cdot 1 = 10, \quad \frac{\partial g}{\partial y} = 2(x + y^2) \cdot 2y = 40$$

3.3 The picture: slopes of two slices

Key point: Fixing x_2 cuts the surface along a curve in the x_1 direction; $\partial f / \partial x_1$ is the slope of its tangent line.

Figure 2 shows the surface of our bowl. Fixing $x_2 = 1$ cuts it along the orange curve, a function of x_1 alone. Its tangent line at $(1, 1)$ has slope 3: that is $\partial f / \partial x_1$. Fixing $x_1 = 1$ instead gives the green curve, whose tangent line has slope 5: that is $\partial f / \partial x_2$.

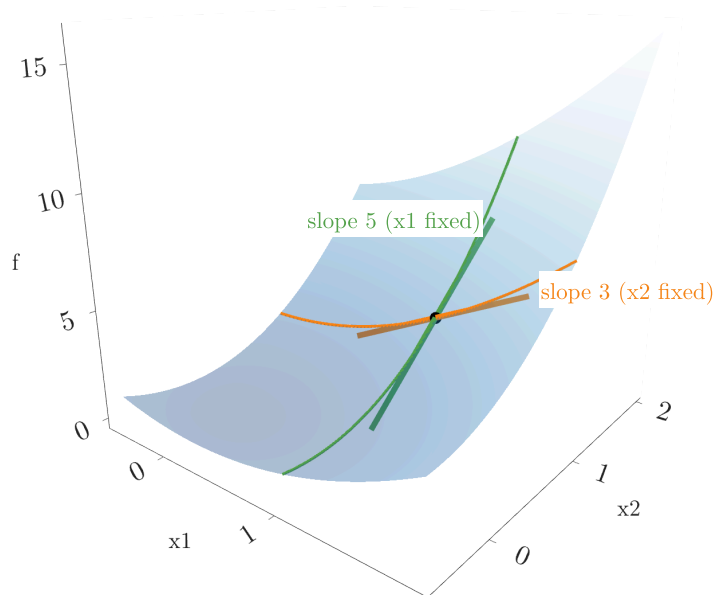


Figure 2: The surface $f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$ cut along $x_2 = 1$ (orange) and along $x_1 = 1$ (green); the thick lines are the tangent lines at $(1, 1)$

The two tangent lines together span a flat plane that touches the surface at $(1, 1)$, the tangent plane. The [Hessian and multivariate Taylor Note](#) uses it to approximate the surface.

4. The gradient

Key point: The gradient collects all partial derivatives in one row vector; as an arrow, it points in the direction of steepest ascent and is perpendicular to the contour lines.

4.1 Collecting the partial derivatives

Key point: One partial derivative per input, side by side: a $1 \times n$ row vector.

An earlier Note on descent ([gradient descent Note](#)) defined the **gradient** as the vector of partial derivatives of the loss, pointing in the direction of steepest increase. We write it as a row:

1. **In words:** list the partial derivatives in the order of the inputs.
2. **Formula:**

$$\nabla_{\mathbf{x}} f = \frac{df}{d\mathbf{x}} = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \cdots \quad \frac{\partial f}{\partial x_n} \right] \in \mathbb{R}^{1 \times n}$$

∇ is read “nabla” or “grad”.

3. **Example:** for our bowl,

$$\nabla f = \begin{bmatrix} 2x_1 + x_2 & x_1 + 4x_2 \end{bmatrix}, \quad \nabla f(1, 1) = \begin{bmatrix} 3 & 5 \end{bmatrix}, \quad \nabla f(2, -1) = \begin{bmatrix} 3 & -2 \end{bmatrix}$$

Row or column? Many texts, and the update rules of the [gradient descent Note](#), treat the gradient as a column vector like the parameters. The book we follow uses a row: then the chain rule becomes ordinary matrix multiplication with no transposes (Section 6), and the next Note’s Jacobian follows naturally. The numbers are the same either way; only the shape differs. In NumPy both are plain 1-D arrays, so code rarely notices.

4.2 The gradient as an arrow on the contour map

Key point: Drawn at its point, the gradient arrow points the way f rises fastest, crosses the contour line at a right angle, and is longer where the surface is steeper.

Figure 1 draws the gradient as an arrow at many points of the contour map. Three things are visible:

- **It points uphill.** Every arrow points away from the minimum, towards higher contour lines.
- **It crosses contour lines at right angles.** Along a contour line f does not change at all, so the steepest direction is straight across it.
- **Its length is the steepness.** Arrows are long where contour lines are packed close together, and short near the flat bottom.

At $(1, 1)$ the gradient $[3, 5]$ (orange) points up and to the right. Its opposite, $[-3, -5]$ (green), points into the bowl: the direction [gradient descent](#) steps in, $\mathbf{x}_{\text{new}} = \mathbf{x} - \eta \nabla f(\mathbf{x})$.

Extra: The slope in any direction follows from the gradient. For a unit vector $\mathbf{u}$ (length 1), the **directional derivative** is the dot product $\nabla f \cdot \mathbf{u}$. At $(1, 1)$:

- along $\mathbf{u} = [1, 0]$ it is 3, the partial derivative $\partial f / \partial x_1$;
- along the gradient itself, $\mathbf{u} = [3, 5] / \sqrt{34}$, it is $\sqrt{34} = 5.83$, the largest possible;
- along $\mathbf{u} = [5, -3] / \sqrt{34}$, at right angles to the gradient, it is $(15 - 15) / \sqrt{34} = 0$: we walk along the contour line.

A dot product with a unit vector is largest when the two point the same way (see the [dot product and cosine similarity Note](#)). That is why the gradient is the direction of steepest ascent.

5. Rules for gradients

Key point: Sum, product and chain rules still hold with vectors, but the factors are now vectors and matrices, so their order matters.

The rules of the one-variable case carry over. For functions $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$:

$$\text{Sum rule: } \frac{\partial}{\partial \mathbf{x}} (f(\mathbf{x}) + g(\mathbf{x})) = \frac{\partial f}{\partial \mathbf{x}} + \frac{\partial g}{\partial \mathbf{x}}$$

$$\text{Product rule: } \frac{\partial}{\partial \mathbf{x}} (f(\mathbf{x}) g(\mathbf{x})) = \frac{\partial f}{\partial \mathbf{x}} g(\mathbf{x}) + f(\mathbf{x}) \frac{\partial g}{\partial \mathbf{x}}$$

$$\text{Chain rule: } \frac{\partial}{\partial \mathbf{x}} g(f(\mathbf{x})) = \frac{\partial g}{\partial f} \frac{\partial f}{\partial \mathbf{x}}$$

With vectors and matrices, AB and BA are different things (see the [matrix multiplication as composition Note](#)). So we keep the order written above and check that the shapes fit.

A building block we need: the gradient of a dot product with a fixed vector $\mathbf{a}$. Since $\mathbf{a}^\top \mathbf{x} = a_1 x_1 + a_2 x_2$, the partial derivatives are a_1 and a_2 :

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{a}^\top \mathbf{x} = \mathbf{a}^\top$$

1. **In words (product rule):** differentiate one factor at a time, keep the other, add.
2. **Formula:** for $f(\mathbf{x}) = (\mathbf{a}^\top \mathbf{x})(\mathbf{b}^\top \mathbf{x})$,

$$\nabla f = (\mathbf{b}^\top \mathbf{x}) \mathbf{a}^\top + (\mathbf{a}^\top \mathbf{x}) \mathbf{b}^\top$$

3. **Example:** $\mathbf{a} = [1, 2]$, $\mathbf{b} = [3, -1]$, $\mathbf{x} = [1, 1]$. Then $\mathbf{a}^\top \mathbf{x} = 3$ and $\mathbf{b}^\top \mathbf{x} = 2$:

$$\nabla f = 2 [1, 2] + 3 [3, -1] = [11, 1]$$

Check by multiplying out: $f = 3x_1^2 + 5x_1x_2 - 2x_2^2$ has partial derivatives $6x_1 + 5x_2 = 11$ and $5x_1 - 4x_2 = 1$.

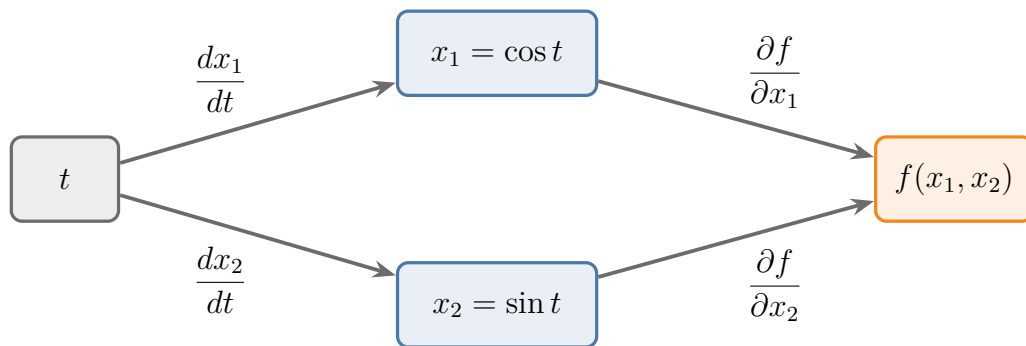
6. The chain rule with several variables

Key point: When an input affects f through several intermediate variables, we multiply along each path and add up the paths; written with gradients, this is a matrix product.

6.1 One outer input, two paths

Key point: The rate of change of f with t is the sum over both paths of (slope of f along the path) times (rate of the intermediate variable): a row vector times a column vector.

Suppose x_1 and x_2 are themselves functions of one variable t , for example a point moving around a circle: $x_1 = \cos t$, $x_2 = \sin t$. How fast does $f(x_1, x_2)$ change as t changes? A change in t reaches f along two paths (Figure 3).



$$\frac{df}{dt} = \underbrace{\frac{\partial f}{\partial x_1} \frac{dx_1}{dt}}_{\text{path through } x_1} + \underbrace{\frac{\partial f}{\partial x_2} \frac{dx_2}{dt}}_{\text{path through } x_2}$$

Figure 3: The multivariate chain rule: multiply the derivatives along each path from t to f , then add the paths

1. **In words:** for each intermediate variable, multiply “how f responds to it” by “how it responds to t ”; add the results.
2. **Formula:**

$$\frac{df}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \frac{\partial f}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial f}{\partial x_2} \frac{dx_2}{dt}$$

3. **Example:** our bowl on the circle. At $t = \pi/2$ the point is $(0, 1)$, where $\nabla f = [0 + 1, 0 + 4] = [1, 4]$. The point moves with velocity $[-\sin t, \cos t] = [-1, 0]$:

$$\frac{df}{dt} = 1 \cdot (-1) + 4 \cdot 0 = -1$$

At $t = 0$ the point is $(1, 0)$, $\nabla f = [2, 1]$, velocity $[0, 1]$, so $df/dt = 1$.

The first form is a 1×2 row times a 2×1 column, giving a 1×1 number. This is where the row-vector convention pays off: the gradient sits on the left and the shapes fit with no transposing.

6.2 Two outer inputs: a matrix of partial derivatives

Key point: With inputs s and t , the inner derivatives fill a 2×2 matrix, and the chain rule becomes (row gradient) times (matrix).

Now let x_1 and x_2 depend on two variables s and t . Each of $\partial f/\partial s$ and $\partial f/\partial t$ follows the path rule:

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial s}, \quad \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

1. **In words:** both equations at once are one row vector times one matrix.
2. **Formula:**

$$\frac{df}{d(s, t)} = \underbrace{\begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix}}_{\partial f / \partial \mathbf{x}} \underbrace{\begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}}_{\partial \mathbf{x} / \partial (s, t)}$$

3. **Example:** our bowl with $x_1 = s + t$ and $x_2 = st$, at $s = 1, t = 2$. Then $x_1 = 3, x_2 = 2$, so $\nabla f = [2 \cdot 3 + 2, 3 + 4 \cdot 2] = [8, 11]$. The inner derivatives are $\partial x_1 / \partial s = 1, \partial x_1 / \partial t = 1, \partial x_2 / \partial s = t = 2, \partial x_2 / \partial t = s = 1$:

$$\begin{bmatrix} 8 & 11 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 8 + 22 & 8 + 11 \end{bmatrix} = \begin{bmatrix} 30 & 19 \end{bmatrix}$$

The matrix in the middle is the first example of a **Jacobian**, the subject of the [Jacobian Note](#). The chain rule as a product of matrices mirrors the [matrix multiplication as composition Note](#): doing one function after another multiplies their matrices.

The notation also reads like fractions: $\partial \mathbf{x}$ appears “below” in the first factor and “above” in the second, and “cancels” to leave $\partial f / \partial (s, t)$. This is a memory aid only; partial derivatives are not really fractions.

7. Checking a gradient numerically

Key point: Nudge each input by a small h and compare the finite-difference slope with the formula; a tiny relative error means the formula and code are probably right.

Hand-derived gradients are easy to get wrong, and a wrong gradient makes training fail quietly. The definition of the partial derivative gives a simple test, called **gradient checking**:

1. **In words:** estimate each partial derivative with a small step h (say 10^{-4}), then compare the whole estimated gradient with the formula’s.
2. **Formula:** with d_i^h the finite-difference estimate and d_i the formula’s value for input i ,

$$\text{relative error} = \sqrt{\frac{\sum_i (d_i^h - d_i)^2}{\sum_i (d_i^h + d_i)^2}}$$

A value below about 10^{-6} means the formula is very probably correct.

3. **Example:** the squared-error loss $L(m, b) = \sum (y_i - mx_i - b)^2$ of the [gradient descent Note](#), on three points $x = [1, 2, 3]$, $y = [2, 4, 5]$, at $m = 1, b = 0$. The residuals are $[1, 2, 2]$, so the formulas give

$$\frac{\partial L}{\partial m} = -2 \sum r_i x_i = -2(1 + 4 + 6) = -22, \quad \frac{\partial L}{\partial b} = -2 \sum r_i = -10$$

Central differences with $h = 10^{-4}$ give $[-22.0000, -10.0000]$, a relative error of about 10^{-13} .

Python:

```
import numpy as np

x = np.array([1.0, 2.0, 3.0])
y = np.array([2.0, 4.0, 5.0])

def loss(p):
    m, b = p
    return np.sum((y - m * x - b) ** 2)

def grad(p):
    r = y - p[0] * x - p[1]
    return np.array([-2 * np.sum(r * x), -2 * np.sum(r)])

p, h = np.array([1.0, 0.0]), 1e-4
num = np.array([(loss(p + h * e) - loss(p - h * e)) / (2 * h)
                 for e in np.eye(2)])
rel = np.linalg.norm(num - grad(p)) / np.linalg.norm(num + grad(p))
num, grad(p), rel      # [-22, -10], [-22, -10], about 2.5e-13
np.eye(2) gives the rows [1, 0] and [0, 1], so p + h * e nudges one parameter at a time.
```

Deep learning libraries compute gradients automatically, but gradient checking is still how those libraries, and any custom loss, are tested.

8. Summary

Object	Shape	Example for $f = x_1^2 + x_1x_2 + 2x_2^2$ at $(1, 1)$
Function value $f(\mathbf{x})$	number	4
Partial derivative $\partial f/\partial x_i$	number	3 and 5
Gradient ∇f	$1 \times n$ row	$[3, 5]$
Chain rule $\partial f/\partial(s, t)$	$(1 \times 2)(2 \times 2) = 1 \times 2$	$[30, 19]$ at $s = 1, t = 2$

- A partial derivative moves one input and holds the others fixed; it is the slope of one slice of the surface.
- The gradient collects all partial derivatives; it points straight uphill, at right angles to the contour lines, and gradient descent steps against it.
- Sum, product and chain rules carry over to vectors, with the order of factors kept.
- The multivariate chain rule adds the products along every path; with gradients as rows, it is matrix multiplication.
- Gradient checking compares a formula with finite differences.

9. Key terms

Term	Meaning
Function of several variables	$f : \mathbb{R}^n \rightarrow \mathbb{R}$: a vector of n numbers in, one number out
Gradient as a row vector	$\nabla f = [\partial f/\partial x_1, \dots, \partial f/\partial x_n] \in \mathbb{R}^{1 \times n}$, the convention that makes the chain rule a matrix product
Nabla (∇)	The symbol for the gradient
Directional derivative	The slope of f along a unit vector $\mathbf{u}$: $\nabla f \cdot \mathbf{u}$
Steepest ascent	The direction in which f increases fastest: the direction of the gradient
Multivariate chain rule	The derivative through intermediate variables: multiply along each path and add the paths; a row gradient times a matrix of inner derivatives
Gradient checking	Testing a gradient formula against finite-difference estimates, using the relative error